QUANT

Venn Diagrams

Assignment 01

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Venn Diagram Online Assignment -01

1. In a class 10 boys play only Khokho, 20 boys play only Kabaddi and 5 boys play both Khokho & Kabaddi. Find the total number of boys in the class.

1. 25 2. 35 3. 42 4. Data inadequate

2. In a club, each member drinks coffee or tea. 19 members in the club drink both coffee and tea, 11 members only coffee and 31 members drink only tea. Find the total number of persons in the club.

1. 31 2. 61 3. 71 4. 51

3. In a colour printer, there are three cartridges containing cyan, magenta and yellow colors. A picture of 25 different squares of the same size (each containing only one colour combination) is to be printed. Of these, six require cyan and magenta, 10 require magenta and yellow. No box requires cyan and yellow. Two require cyan only, 1 requires magenta only and 13 require yellow. Find the percentage of area, which is left white.

1. 20 % 2. 3 % 3. 21 % 4. 12 %

4. In the above example, one of the cartridges was dry and hence that colour was not printed. If the area which is left white is now doubled, which colour was dry?

1. Yellow 2. Magenta 3. Cyan 4. Data inadequate

5. In a class, 31 boys fail both in English and Mathematics. 29 boys fail only in Mathematics. 17 boys fail only in English. If the percentage of boys passing in Mathematics is 75 % of the percentage of boys passing in English, find the class strength.

1. 96 2. 161 3. 115 4. None of these

6. In the above question, how many students pass in only one of the subjects?

1. 65 2. 55 3. 46 4. None of these

7. In a school, 49 boys play both hockey and volleyball. 76 boys play only hockey. 65 boys play volleyball. Find the percentage of boys playing volleyball, if the class strength is four times the number of boys playing hockey.

1. 33.33 % 2. 13 % 3. 23 % 4. None of these

8. In a girls' school, 128 girls play skipping, 102 play throw-ball. Out of these 30 girls play both throw-ball and skipping. Find the percentage of girls not playing any of these games, if the number of girls playing less than two games is nine times the number of girls playing two games.

1. 25 % 2. 40 % 3. 33.33 % 4. 18.18 %

9. A shopkeeper observes that 125 customers like to use red, but won't mind using blue colored pencils; 115 customers like to use only red colored pencils; 210 customers like to use only blue colored pencils. If he runs out of supply of red pencils, find the percentage of sales lost, assuming that each customer buys the same number of pencils.

1. 23 % 2. 27.27 % 3. 33.33 % 4. 25.55 %

10. In the above example, the shopkeeper realizes that blue pencils wear out faster and hence, eight blue pencils are used to do the work equivalent of five red pencils. Find the net loss in sales now.

1. 6 % 2. $^{80}/_{9}$ % 3. 12.5 % 4. $^{120}/_{49}$ %

Answer Key & Explanations

Sr. No.	Key	Explanation
1	4	10 + 20 + 5 = 35 seems to be the answer, but it is wrong, as the question never specifies that everybody plays one of these two games. There could be students in the class, who play none of these two games.
2	2	The total no. of persons will become the sum of all three i.e. 19 + 11 + 31 = 61.
3	4	You can make the diagram or you can think logically. There are 13 who require yellow, this means the whole of yellow colour is included. Then 6 require cyan and magenta. As no box requires cyan and yellow. Thus these six require cyan and magenta only. Then two require cyan only and 1 requires magenta only. Thus there are 13 + 6 + 2 + 1 = 22 squares. The remaining three boxes are white. This implies $100 \times \frac{3}{25} = 12\%$ of the boxes are left white.
4	1	Because the white area is doubled, this implies this colored must have been used single only on as many squares as there were unpainted squares. Just see which colors could have been used on three squares. It cannot be cyan and magenta as their no. is specified in the question, the only remaining color is yellow which alone was used on three squares. Thus it should be the answer.
5	1	As the values given 12 extra students are fail in Mathematics or 12 extra students are pass in English. The question states pass in Mathematics is 75% of pass in English. Now the 25% extra in English must be 12 in no. The students who are pass in English are $100 \times \frac{12}{25} = 48$. But out of these 29 students are pass in English only (as it is the no. given to be fail in Mathematics only). Now the total strength of the class is 31 + 29 + 17 + 19 = 96.
6	3	29 + 17 = 46 (students who are fail in Mathematics only, means they are pass in English only and vice versa).
7	2	The boys playing hockey are 49 + 76 = 125. The class strength is four times of that i.e. 125 × 4 = 500. Out of these the no. of boys playing volleyball is 65 (mind the 49 who play both are already included in this, because the word only is not written with it). The percentage is thus $100 \times \frac{65}{500} = 13\%$.
8	3	The no. of girls playing two games 30. Number of girls playing less than two games 30 × 9 = 270. Girls playing one or two of these games are 128 + 102 – 30 = 200. Girls playing none of these two games are 270 + 30 – 200 = 100. The percentage is $100 \times \frac{100}{300} = 33.33\%$
9	4	Assume each of the customers wants one pencil. His supposed sales should have been 125 +115 + 210 = 450. Out of which 115 would not buy pencils from him. The percentage of sales lost is $100 \times \frac{115}{450} = 25.55\%$
10	2	Now those 125 customers would buy $125 \times \frac{8}{5}$ = 200 pencils. Now his sale will be 200 + 210 = 410. The sales loss is now 45 0 – 410 = 40. The percentage loss in sales is $100 \times \frac{40}{450} = \frac{80}{9}\%$